# How Should a Penalized VAR Be Tuned to Measure Connectedness?*

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**Abstract**

Connectedness indexes built from elastic-net VARs depend on a penalty that is usually chosen by ten-fold cross-validation with folds assigned at random. With persistent series, an observation held out for validation keeps its neighbours in the training sample, so cross-validation overstates how well a dense model predicts. In a Monte Carlo with a known sparse network, I compare seven ways of assigning observations to folds, two points on the cross-validation curve, information criteria and a plug-in penalty. When the median first autocorrelation is 0.88, random folds keep three times as many cross-lag coefficients as the best predicting penalty, add 35 percent to the error in pairwise connectedness and raise the index by about nine points. Folds separated by a gap of six to twelve observations, or a rolling origin, stay within 5 percent of the best penalty at every persistence level. Choosing the one-standard-error penalty is no remedy. Whatever the rule, about $100(N-1)/(T+N-1)$ points of a reported index come from noise in the estimated shock covariance alone.

JEL: C32, C52, C53, C15 Keywords: connectedness, elastic-net VAR, cross-validation, persistence, forecast error variance decomposition, Monte Carlo

*Preliminary draft. Comments welcome. Replication code is available from the author.

# 1 Introduction

The connectedness index of Diebold and Yilmaz (2012, 2014) summarises how much of the forecast error of each series in a system is due to shocks in the other series. With many series the underlying VAR has more coefficients than a short sample can estimate, so applied work estimates it with the lasso or the elastic net. The penalty that decides how many coefficients survive is chosen by ten-fold cross-validation. Demirer et al. (2018), Diebold et al. (2017), Bostancı and Yılmaz (2020), Gross and Siklos (2020), Gabauer et al. (2024), Jiménez-Varón and Knight (2025) and Manso (2026) all say so, Greenwood-Nimmo and Tarassow (2022) use five folds, and none of them says how the observations are assigned to folds. The cross-validation routine of Friedman et al. (2010) assigns them at random by default, and Brunner and Hipp (2023) are the only authors I have found who state that their folds are blocked.[1]

Random folds were made for independent observations. The series these indexes are built from are persistent. House prices, volatilities and credit spreads move slowly, and an observation held out for validation is almost the same as the ones before and after it, which stay in the training sample. Cross-validation then rewards a model for fitting the neighbours of the held-out observation, so it overstates how well a model with many coefficients predicts and picks too small a penalty. A smaller penalty keeps more cross-lag coefficients, and every coefficient kept is a link in the network the index summarises. Bergmeir et al. (2018) show that random folds are valid for an autoregression only if the fitted model leaves no autocorrelation in its residuals, which is not what a heavily penalized model does with persistent series.

This paper asks which rule for choosing the penalty recovers the connectedness network when the series are persistent. I answer it with a Monte Carlo in which the truth is known exactly. The data come from a sparse VAR(2) with 50 or 100 series, 2.5 cross links per equation and samples of 120, 300 or 600 observations. I hold the network fixed and vary the persistence of the series, from a median first autocorrelation of 0.24 to 0.88. The last is the persistence of monthly state house prices. In each simulated sample I estimate the elastic-net VAR the way applied work does, equation by equation over one path of penalties, and let every rule choose from that path. The rules are cross-validation with random, interleaved and blocked folds, hv-block cross-validation with gaps of 2, 6 and 12 observations, a rolling origin, the minimum and the one-standard-error point of each cross-validation curve, BIC, EBIC and a plug-in penalty. Because the VAR is known, I also know the true out-of-sample error of every fit and the true connectedness table at every horizon. This gives two benchmarks that no estimator could use in an application, because they use the truth. Following the statistics literature I call them oracles. The prediction oracle is the penalty with the lowest true out-of-sample error, and the network oracle is the penalty with the lowest true error in the connectedness table. Every feasible rule is judged by how far it falls from them.

Random folds fail exactly where the series are persistent. At the highest persistence, random ten-fold cross-validation estimates an out-of-sample error $-1.26$ shock variances below the true

[1] The routine is `cv.glmnet` in the R package glmnet. The ConnectednessApproach package (Gabauer, 2025), which implements the elastic-net VAR of Gabauer et al. (2024), calls it without a fold vector, so its folds are random as well.

one, chooses a penalty 10.9 steps of the path below the best predicting penalty, about three times smaller, and keeps 30 cross-lag coefficients per equation where the best predicting penalty keeps 10. The error in pairwise connectedness is 35 percent larger than at the best penalty for the network, and the index is 27.2 points above the truth instead of 17.2. The out-of-sample error is 126 percent of the shock variance above the irreducible error, against 51 percent at the best predicting penalty. Interleaved folds, which assign every tenth observation to the same fold, do no better. At low persistence none of this matters much. Random folds then add 1 percent to the network error.

Rules that keep the validation observations away from the training observations do well at every persistence level. hv-block cross-validation with a gap of 6 or 12 observations and a rolling origin stay within 5 percent of the best penalty for the network, and they have the three best mean ranks among the 18 feasible rules over the 18 cells of the grid. Blocked folds without a gap come close, but at the highest persistence they still overfit by 1.7 steps. The one-standard-error rule is no remedy. It moves every rule toward heavier penalties, which partly undoes the leak of random folds but over-shrinks every rule that does not leak. With a rolling origin, it adds 42 percent to the network error at the highest persistence. EBIC and the plug-in penalty also over-shrink. BIC breaks down when an equation has more regressors than observations, but BIC with a lower bound on the penalty, as Hecq et al. (2023) recommend, is within 2 percent of the network oracle at every persistence level and needs no cross-validation at all. Predicting well and recovering the network turn out to be close goals. The best predicting penalty is within 1 percent of the best penalty for the network. These rankings hold at every horizon from two steps ahead on and reverse only one step ahead. The one-step table depends on the residual correlations alone, which the denser fits of the leaky rules estimate slightly better, so a one-step table is the wrong place to choose a tuning rule.

The second result does not depend on the rule. Even with the true coefficients, the index is not zero when the network is empty, because the shock covariance is estimated. I show that with an empty network this is about $100(N-1)/(T+N-1)$ points, which is 28 points for 48 series and $T = 120$, the size of a ten-year window of monthly data, and the simulations match it within 0.3 points. With links, the tuning rules move the index around this number. They cannot remove it.

Brunner and Hipp (2023) are the first to score penalized VARs by how well they recover a known connectedness table. They set the penalty at its best value, with persistence of at most 0.5, and in their application they use blocked folds. I ask which penalty a researcher can actually choose gets close to their benchmark, and how the answer changes with persistence. Hecq et al. (2023) compare information criteria, a plug-in rule and expanding-window cross-validation for Granger-causality tests in large VARs, and find that BIC with a lower bound on the penalty and the plug-in rule work best. They do not compare fold assignments and their target is the size of a test, not a network. The case for leaving a gap between training and validation observations goes back to Burman et al. (1994) and Racine (2000), and Bergmeir et al. (2018) and Lusompa (2025) show when random folds are and are not valid for autoregressions. These papers judge cross-validation by prediction, and Nicholson et al. (2020) warn that the penalty that predicts best need not recover structure. I find that, for the connectedness network, the warning has little bite once the folds do not leak. Finally,

Brunner and Hipp (2023), Greenwood-Nimmo et al. (2024) and Bonaccolto et al. (2026) show that estimated connectedness is biased upward. I separate the part of this bias that comes from the shock covariance alone and give its size, which tells a reader how much of a reported index no tuning rule can remove.

## 2 Estimator and tuning rules

Applied work builds the index in three steps. A penalized VAR supplies the lag coefficients and the residual covariance, a forecast error variance decomposition turns them into a table of pairwise shares, and a tuning rule decides the penalty that the first step uses. The Monte Carlo keeps the first two steps exactly as applied work carries them out and varies only the third, so this section describes all three in that order.

### 2.1 Elastic-net VAR

Let $y_t$ be a vector of $N$ series observed for $T$ periods. The VAR(2) is

$$y_t = A_1 y_{t-1} + A_2 y_{t-2} + e_t, \qquad e_t \sim N(0, \Sigma), \tag{1}$$

where $A_1$ and $A_2$ are $N \times N$ coefficient matrices and $\Sigma$ is the covariance of the shocks. The series have mean zero, so the process has no intercept. The estimator fits one, as applied work does, and because the intercept is not penalised a nonzero mean would be absorbed exactly and change nothing below. Each equation has $2N$ slopes and an intercept, so with $N = 100$ and $T = 120$ there are 201 coefficients to estimate from 120 observations. I estimate each equation by the elastic net (Zou and Hastie, 2005). For equation $i$, with $x_t = (y'_{t-1}, y'_{t-2})'$,

$$(\hat{a}_i, \hat{\beta}_i) = \arg\min_{a,\beta} \; \frac{1}{2T} \sum_t \left(y_{it} - a - x'_t\beta\right)^2 + \lambda_i \left(\alpha\|\beta\|_1 + \frac{1-\alpha}{2}\|\beta\|_2^2\right), \tag{2}$$

with $\alpha = 0.5$, the regressors standardised and every lag, own or cross, penalised alike. These are the choices applied work makes, and I hold them fixed. The penalty $\lambda_i$ runs over a path of 100 values spaced evenly in logs, from the smallest value that sets every slope to zero down to $10^{-4}$ times that value, so one step of the path changes the penalty by about 10 percent. A tuning rule picks one step for each equation.

### 2.2 Connectedness table

Write $\Psi_h$ for the moving-average coefficients of the fitted VAR, so that $\Psi_0 = I$ and $\Psi_h = A_1\Psi_{h-1} + A_2\Psi_{h-2}$. The generalized forecast error variance decomposition of Koop et al. (1996) and Pesaran and Shin (1998) gives the share of the $H$-step forecast error of series $i$ that is due to shocks in series

$j$,

$$\theta_{ij}(H) = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} \left(e_i' \Psi_h \Sigma e_j\right)^2}{\sum_{h=0}^{H-1} e_i' \Psi_h \Sigma \Psi_h' e_i}, \tag{3}$$

where $e_i$ is the $i$-th unit vector and $\sigma_{jj}$ is the variance of shock $j$. Because the shocks are correlated, the shares of a row do not sum to one, so Diebold and Yilmaz (2012) divide each by its row sum,

$$C_{ij}(H) = 100 \, \frac{\theta_{ij}(H)}{\sum_{k=1}^{N} \theta_{ik}(H)}, \qquad \mathrm{DYCI}(H) = \frac{1}{N} \sum_{i=1}^{N} \sum_{j \neq i} C_{ij}(H). \tag{4}$$

$C_{ij}$ is the pairwise connectedness from $j$ to $i$ in percentage points, and the index DYCI is the average share of a series' forecast error that comes from the others. The estimated table replaces $A_1$, $A_2$ and $\Sigma$ with the fitted coefficients and the covariance of the fitted residuals. My main horizon is $H = 12$ steps ahead, a year for monthly data.

### 2.3 Tuning rules

Every cross-validation rule splits the sample into training and validation observations, fits the path on the training observations and computes the mean squared error of each step of the path on the validation observations. The rules differ only in which observations they hold out.

- **Random ten-fold.** Each observation is assigned to one of ten folds at random, the default of Friedman et al. (2010).
- **Interleaved ten-fold.** Observation $t$ goes to fold $t$ modulo 10, so the validation observations are spread evenly over the sample.
- **Blocked ten-fold.** The sample is cut into ten consecutive blocks.
- **hv-block** (Burman et al., 1994; Racine, 2000). The sample is cut into twelve consecutive blocks, and the $h$ observations on each side of the validation block are dropped from training, with $h = 2$, 6 or 12.
- **Rolling origin.** The last half of the sample is cut into five blocks of a tenth of the sample each, and each block is predicted from all observations before it.

Each curve gives two choices, the step with the lowest error ($\lambda_{\min}$) and the largest penalty whose error is within one standard error of that minimum ($\lambda_{1se}$), where the standard error is taken across folds. Three rules do not hold out data. BIC (Schwarz, 1978) picks, for each equation, the step that minimises $\log(\mathrm{RSS}/T) + \mathrm{df} \log T/T$, where df is the number of nonzero slopes. EBIC (Chen and Chen, 2008) adds $\mathrm{df} \log p/T$ with $p = 2N$ candidate regressors. The plug-in penalty follows Belloni et al. (2012) with an iterated estimate of the residual variance. Every rule chooses from the same path, so the rules differ only in the step they pick.

# 3 Monte Carlo simulation

The simulation holds a sparse network fixed and varies the persistence of the series around it, because persistence is what random folds are blind to. Everything the rules are judged by, the forecast error of a fit and the connectedness table, is known exactly, and two oracles mark the best that any rule could do.

## 3.1 Designs

The data come from the VAR(2) in Equation (1) with

$$A_1 = 0.7\,R + 0.7\,\kappa\,(S \circ G), \qquad A_2 = 0.3\,R + 0.3\,\kappa\,(S \circ G), \qquad R = \mathrm{diag}(\rho_1, \ldots, \rho_N), \tag{5}$$

where $\rho_i$ is the sum of the own-lag coefficients of series $i$, $G$ is a 0/1 matrix of cross links, $S$ gives each link a sign and $\kappa$ is the size of every link. Each off-diagonal entry of $G$ is a link with probability $2.5/N$, so every equation has 2.5 cross links on average whatever $N$ is, and 85 percent of the links are positive. The shocks have unit variances and correlations $\Sigma_{ij} = 0.3^{|i-j|}$.

Persistence is the dimension I vary. The own-lag sums are $\rho_i = \bar{\rho} + \delta u_i$ with $u_i$ uniform on $[-1, 1]$, and Table 1 shows the four levels. At each level I solve for the link size $\kappa$ that makes the lagged part of the true index 5 points at $H = 12$. The correlated shocks add 16.3 points at every level, so the true index is 21.3 points at every level and only persistence changes. The network is drawn once per cell and held fixed across replications, and only the shocks are drawn again. The control level gives every series the same persistence and is simulated at $N = 50$ and $T = 600$ only. The two other dimensions are the number of series, $N \in \{50, 100\}$, and the length of the sample, $T \in \{120, 300, 600\}$ observations. With $N = 100$ and $T = 120$ an equation has more regressors than observations. A cell is one combination of family, persistence level, $N$ and $T$. The linked family has 18 cells, three persistence levels by two values of $N$ by three of $T$, plus the equal-persistence cell, and every cell is simulated 100 times.

**Table 1:** Persistence levels of the simulated VARs

| | | First autocorrelation | | Largest root | | |
|---|---|---|---|---|---|---|
| Level | Own-lag sum $\bar{\rho} \pm \delta$ | Median | 10th–90th pct. | $N = 50$ | $N = 100$ | Link size $\kappa$ |
| Low | $0.3 \pm 0.20$ | 0.24 | 0.12–0.35 | 0.64 | 0.68 | 0.159 |
| High | $0.7 \pm 0.20$ | 0.61 | 0.47–0.77 | 0.92 | 0.92 | 0.072 |
| Very high | $0.9 \pm 0.05$ | 0.88 | 0.83–0.92 | 0.97 | 0.98 | 0.038 |
| Equal persistence | 0.7 | 0.64 | 0.62–0.66 | 0.85 | – | 0.068 |

*Note:* Linked designs. First autocorrelation of the simulated series at $N = 50$, across series. Largest root is the largest modulus of the companion matrix. The link size $\kappa$ is solved so that the lagged part of the true connectedness index at $H = 12$ is 5 points; it is shown for $N = 50$. The true index at $H = 12$ is 21.3 points at every level, 16.3 of which come from the correlated shocks. The equal-persistence level is simulated at $N = 50$ and $T = 600$ only.

A second family of designs has no links at all ($\kappa = 0$) and uncorrelated shocks, so the true index is zero at every horizon and every cross-lag coefficient a rule keeps is false. It has the same 19 cells.

### 3.2 Outcomes

Because the VAR is known, the truth is exact. The true table follows from Equations (3) and (4) with the true $A_1$, $A_2$ and $\Sigma$. The true out-of-sample error of any fitted equation is

$$\mathrm{MSE}_i(\hat{a}_i, \hat{\beta}_i) = \Sigma_{ii} + (\beta_i - \hat{\beta}_i)' \, \Gamma_X \, (\beta_i - \hat{\beta}_i) + \hat{a}_i^2, \tag{6}$$

where $\beta_i$ are the true slopes and $\Gamma_X$ is the stationary covariance of $x_t$. The first term is the error no model can avoid. The second is the cost of the estimated coefficients, and the third the cost of the fitted intercept.

I judge each rule by two kinds of outcomes. The first kind asks whether cross-validation does its job. Optimism is the cross-validated error at the chosen step minus the true error at that step, averaged over equations, and a negative value means cross-validation promises more than the fit delivers. I also count how many steps of the path the chosen penalty lies below the penalty that predicts best. The second kind asks what the choice does to the network. The network error is the mean absolute error of the pairwise shares $C_{ij}$ over all $i \neq j$, and the index error is the estimated index minus the true one. For the links themselves I report the share of the true cross-lag coefficients a rule keeps, the share of the zero coefficients it wrongly keeps, and the Matthews correlation between two 0/1 variables, whether a coefficient is kept and whether it is truly nonzero. The true links are rare, 2.5 per equation among $2N$ candidates, so a rule that keeps nothing is right about almost every coefficient and still finds no link. The Matthews correlation is zero in that case and one when the kept and the true links coincide.[2]

### 3.3 Oracles

Every rule in Section 2.3 picks one step from the same path of penalties, using only the sample. An oracle picks a step from that same path, but it is allowed to look at the truth, which no estimator can in an application. It is the best any rule could do with the fits on the path, so the distance between a rule and an oracle measures how much the rule loses by not knowing the truth, and nothing else. I use two oracles, one for prediction and one for the network, so that each goal is judged against its own truth.

The prediction oracle picks, for each equation, the step whose fit has the lowest true out-of-sample error,

$$s_i^P = \arg \min_{s=1,\ldots,100} \; \mathrm{MSE}_i(\hat{a}_i(s), \hat{\beta}_i(s)), \tag{7}$$

where $\hat{a}_i(s)$ and $\hat{\beta}_i(s)$ are the coefficients of equation $i$ at step $s$ and $\mathrm{MSE}_i$ is the true error in Equation (6). Cross-validation tries to estimate exactly this error, so the prediction oracle is what cross-validation would pick if its estimate had no noise and no bias.

The network oracle picks the step that makes the estimated connectedness table closest to the

[2] With $a$ true links kept, $b$ zero coefficients kept, $c$ true links dropped and $d$ zero coefficients dropped, it is $(ad - bc)/\sqrt{(a+b)(a+c)(b+d)(c+d)}$.

true one,

$$s^N = \arg \min_{s=1,\ldots,100} \frac{1}{N(N-1)} \sum_{i \neq j} |\hat{C}_{ij}(12; s) - C_{ij}(12)|, \tag{8}$$

where $\hat{C}_{ij}(12; s)$ is the pairwise connectedness at $H = 12$ when every equation is estimated at step $s$. The table depends on all equations at once, so a separate step for each equation would mean searching over $100^N$ combinations. The network oracle therefore uses one step for all equations. This is a restriction the prediction oracle does not have, and it is why the prediction oracle can come out slightly ahead of the network oracle on the network error.

A third benchmark is not a choice of penalty. It uses the true coefficients and estimates only the covariance of the shocks, from the residuals the true coefficients leave. Whatever error it has comes from the estimated covariance alone, which no penalty can change.

# 4 Results

Random folds fail in a chain. They misjudge the forecast error, the misjudged error picks too small a penalty, and the penalty keeps links that do not exist. The section follows that chain through the network and the index, and then asks what else a researcher could do, shrink more, use an information criterion or wait for a longer sample. It ends with the part of the index that no rule removes and with the one horizon at which the ranking of the rules turns.

## 4.1 Leakage

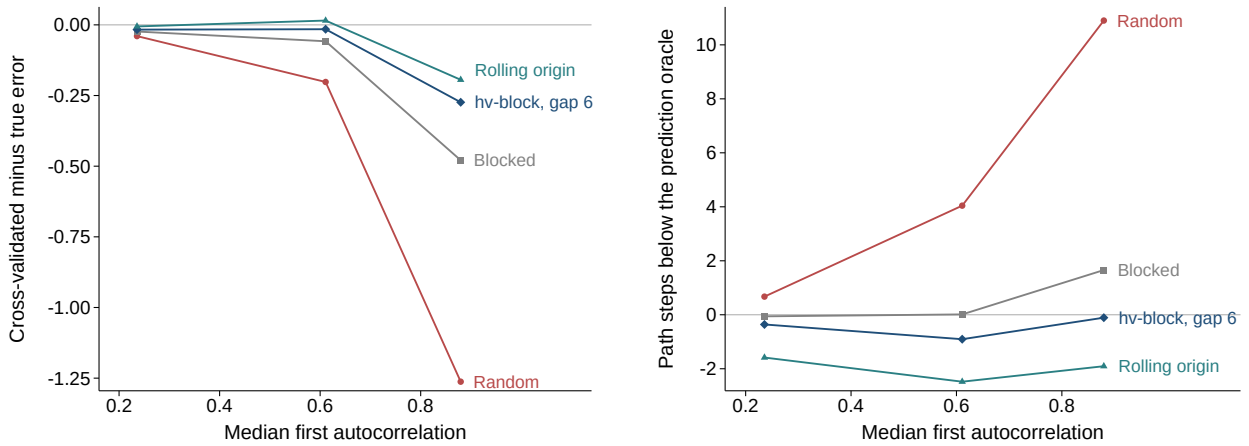
Cross-validation with random folds underestimates the out-of-sample error, and by more the more persistent the series are (Figure 1a).[3] At low persistence the shortfall is $-0.04$ shock variances, a few percent of the irreducible error. At high persistence it is $-0.20$, and at very high persistence it is $-1.26$, more than the variance of the shock itself. A held-out observation of a series with a first autocorrelation of 0.88 is almost a copy of its neighbours, so the fit to the neighbours passes for a forecast. Interleaved folds leak in the same way, which is expected, because every validation observation sits between two training observations. Blocked folds leak only at the edges of each block, and their shortfall is about a third as large. Leaving a gap between training and validation removes most of the rest. With a gap of 12 observations the shortfall is $-0.12$, and with a rolling origin it is $-0.19$.

The optimism shows up in the penalty. Random folds pick a penalty 4.0 steps below the best predicting one at high persistence and 10.9 steps below at very high persistence, which makes the penalty about three times smaller (Figure 1b). Rules with a gap pick penalties at or slightly above the best predicting one. The smaller penalty keeps more cross-lag coefficients.[4] At very high persistence random folds keep 30 per equation, three times as many as the best predicting penalty,

[3] See Table A.1 for all seven cross-validation rules.

[4] See Table A.1, panel C.

while hv-block with a gap of 6 keeps 9. The true VAR has 5, so most of the coefficients random folds keep are links that do not exist.




**(a)** Optimism **(b)** Chosen penalty

**Figure 1:** What cross-validation promises and what it chooses, by persistence

*Note:* Linked designs, $\lambda_{\min}$. Panel (a) shows the cross-validated mean squared error at the chosen step minus the true out-of-sample error at that step, in units of the shock variance. Panel (b) shows how many steps of the penalty path the chosen penalty lies below the prediction oracle's; each step changes the penalty by about 10 percent. The horizontal axis is the median first autocorrelation of the simulated series at the low, high and very high persistence levels. Each point is the mean over the six $N \times T$ cells of a level. Random and Blocked are ten-fold cross-validation with random and consecutive folds.

## 4.2 Network error

The extra links make the estimated network worse (Table 2). At low persistence every cross-validation rule at $\lambda_{\min}$ is within 1 percent of the network oracle, so the choice of folds does not matter. At high persistence random folds add 10 percent to the error in pairwise connectedness and the rules with a gap add 2. At very high persistence random folds add 35 percent and the rules with a gap add 4 to 5. The index moves in the same direction. With random folds it is 27.2 points above the truth at very high persistence, against 18.5 points with hv-block and a gap of 6, so about nine points of the index are created by the leak. The leak also hurts the forecast the model is supposed to deliver. The out-of-sample error of random folds is 126 percent of the shock variance above the irreducible error, twice as much as hv-block's 61 percent. A researcher who uses random folds on persistent data therefore gets a denser network, a higher index and worse forecasts, all at once.

The means hide little. Figure 2 draws the distribution of the excess network error across the 600 replications of each level. At very high persistence the 10th percentile for random folds, 19.6 percent, lies above the 90th percentile for hv-block with a gap of 6, 7.1 percent, and above the 90th for the rolling origin, 10.9 percent, so the distributions do not overlap. At high persistence they barely touch, 5.1 against 3.9 percent. At low persistence the three lie on top of one another, with medians of 0.7 and 0.1 percent.

**Table 2:** Error in the connectedness network by tuning rule and persistence

| | Low | | High | | Very high | | |
|---|---|---|---|---|---|---|---|
| Rule | Network | Index | Network | Index | Network | Index | Mean rank |
| *K-fold and gap cross-validation,* $\lambda_{\min}$ | | | | | | | |
| Random 10-fold | 1 | 13.8 | 10 | 18.0 | 35 | 27.2 | 11.6 |
| Interleaved 10-fold | 1 | 13.9 | 11 | 18.3 | 37 | 27.8 | 12.8 |
| Blocked 10-fold | 0 | 13.5 | 2 | 15.2 | 4 | 18.7 | 6.3 |
| hv-block, gap 2 | 0 | 13.5 | 2 | 15.1 | 4 | 18.5 | 6.1 |
| hv-block, gap 6 | 0 | 13.3 | 2 | 15.1 | 4 | 18.5 | 4.4 |
| hv-block, gap 12 | 0 | 13.2 | 2 | 15.2 | 5 | 19.0 | 3.7 |
| Rolling origin | 0 | 12.9 | 1 | 14.9 | 5 | 18.8 | 4.1 |
| *K-fold and gap cross-validation,* $\lambda_{1se}$ | | | | | | | |
| Random 10-fold | 4 | 12.6 | 5 | 15.3 | 3 | 18.0 | 8.1 |
| Interleaved 10-fold | 4 | 12.6 | 5 | 15.2 | 3 | 17.9 | 7.2 |
| Blocked 10-fold | 4 | 12.6 | 12 | 17.3 | 18 | 22.1 | 12.1 |
| hv-block, gap 2 | 4 | 12.6 | 13 | 17.6 | 19 | 22.7 | 12.7 |
| hv-block, gap 6 | 4 | 12.6 | 14 | 18.0 | 24 | 24.4 | 14.3 |
| hv-block, gap 12 | 4 | 12.7 | 15 | 18.5 | 29 | 26.1 | 15.6 |
| Rolling origin | 5 | 12.8 | 21 | 19.7 | 42 | 28.6 | 17.0 |
| *Information criteria and plug-in penalty* | | | | | | | |
| BIC | 19 | 20.0 | 16 | 21.5 | 14 | 23.6 | 6.9 |
| BIC, bounded | 2 | 12.4 | 2 | 14.9 | 2 | 17.8 | 4.6 |
| EBIC | 5 | 12.9 | 8 | 16.5 | 12 | 20.9 | 10.4 |
| Plug-in | 5 | 12.7 | 12 | 17.5 | 16 | 22.0 | 13.3 |
| *Infeasible benchmarks* | | | | | | | |
| Network oracle | 0 | 12.5 | 0 | 14.2 | 0 | 17.2 | |
| Prediction oracle | −2 | 12.9 | −1 | 14.5 | 1 | 18.0 | |
| True coefficients | −17 | 15.3 | −20 | 15.3 | −26 | 15.2 | |

*Note:* Linked designs, $H = 12$. Network is the mean absolute error of the pairwise shares $C_{ij}$, in percent above the network oracle's in the same cell. Index is the estimated connectedness index minus the true one, in points; the true index is 21.3. Each entry is the mean over the six $N \times T$ cells of a persistence level. Mean rank orders the feasible rules by network error within each of the 18 cells of the three levels (1 is best) and averages the ranks. The network oracle uses one penalty step for all equations, while the prediction oracle chooses a step for each equation, so the prediction oracle can have a lower network error. True coefficients use the true $A_1$ and $A_2$ with the covariance of their own residuals. Monte Carlo standard errors of the network entries are at most 0.2 percent.

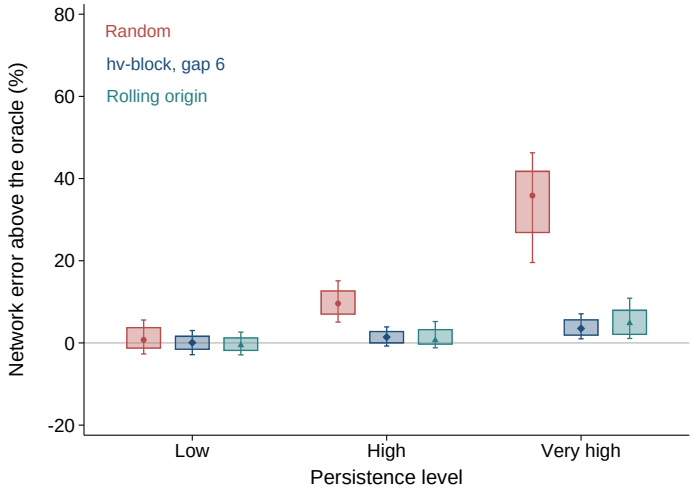



**Figure 2:** The excess network error across replications, by persistence and tuning rule

*Note:* Linked designs, $H = 12$, $\lambda_{\min}$. The network error of a rule in percent above the network oracle's in the same replication, across the 600 replications of a level (100 in each of the six $N \times T$ cells). Each box runs from the 25th to the 75th percentile, the whiskers from the 10th to the 90th, and the mark is the median.

Ranked by network error across the 18 cells, the three rules with the longest separation between training and validation come first. hv-block with a gap of 12 has a mean rank of 3.7, the rolling origin 4.1 and hv-block with a gap of 6 4.4. Random folds at $\lambda_{\min}$ rank 11.6 of 18. Blocked folds and hv-block with a gap of 2 are close behind. At very high persistence they still pick penalties one to two steps too small, but this costs the network little. The leak is not only a matter of the spread of persistence across series. In the control design, where every series has the same persistence, random folds add 9 percent to the network error and the rolling origin 1 percent.

### 4.3 Link recovery

The connectedness table can be recovered without recovering the links. Table 3 reports, for each rule, the share of the true cross-lag coefficients it keeps, the share of the zero coefficients it wrongly keeps and the Matthews correlation between the kept and the true links. No rule finds the network link by link. Even the prediction oracle keeps only 48 percent of the true coefficients at low persistence and 24 percent at very high persistence, and no rule has a Matthews correlation above 0.25. Each link in the design is small, because the lagged part of the true index is only 5 points, so most links are too weak to tell apart from noise in samples of this length. What separates the rules is the false links. At very high persistence random folds keep 20.4 percent of the zero coefficients, against 6.5 percent for hv-block with a gap of 6, and their Matthews correlation falls to 0.08. These false links are what raise the index and the pairwise error. The rules that recover the table best keep about as many true and false links as the oracles do.

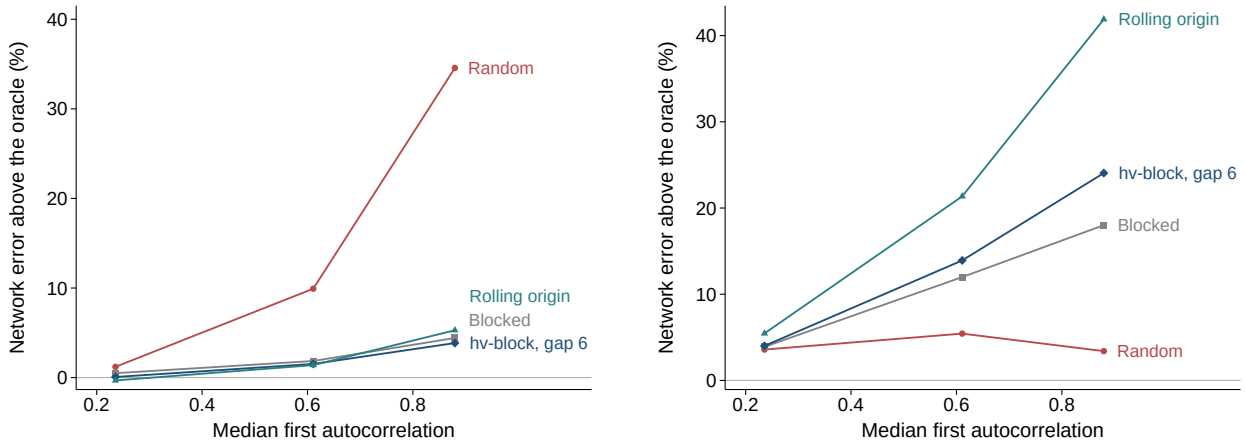
**Table 3:** Recovery of the cross-lag links by tuning rule and persistence

| | Low | | | High | | | Very high | | |
|---|---|---|---|---|---|---|---|---|---|
| Rule | True | False | MCC | True | False | MCC | True | False | MCC |
| *Cross-validation,* $\lambda_{\min}$ | | | | | | | | | |
| Random 10-fold | 45 | 8.7 | 0.22 | 36 | 13.6 | 0.12 | 37 | 20.4 | 0.08 |
| Interleaved 10-fold | 46 | 8.8 | 0.22 | 36 | 14.1 | 0.12 | 38 | 21.2 | 0.08 |
| Blocked 10-fold | 43 | 7.5 | 0.23 | 27 | 7.0 | 0.14 | 24 | 8.0 | 0.11 |
| hv-block, gap 2 | 43 | 7.4 | 0.23 | 26 | 6.6 | 0.14 | 24 | 7.4 | 0.11 |
| hv-block, gap 6 | 42 | 7.1 | 0.23 | 25 | 6.0 | 0.14 | 22 | 6.5 | 0.12 |
| hv-block, gap 12 | 41 | 6.6 | 0.23 | 24 | 5.5 | 0.14 | 21 | 5.8 | 0.12 |
| Rolling origin | 38 | 5.2 | 0.24 | 21 | 4.4 | 0.14 | 19 | 5.4 | 0.12 |
| *Cross-validation,* $\lambda_{1se}$ | | | | | | | | | |
| Random 10-fold, $\lambda_{1se}$ | 11 | 0.4 | 0.21 | 11 | 1.8 | 0.14 | 18 | 5.4 | 0.12 |
| hv-block, gap 6, $\lambda_{1se}$ | 9 | 0.2 | 0.20 | 6 | 0.4 | 0.13 | 10 | 1.8 | 0.12 |
| *Information criteria and plug-in penalty* | | | | | | | | | |
| BIC | 26 | 14.5 | 0.22 | 25 | 16.0 | 0.14 | 29 | 17.0 | 0.12 |
| BIC, bounded | 11 | 0.2 | 0.23 | 11 | 1.2 | 0.15 | 17 | 4.2 | 0.12 |
| EBIC | 4 | 0.7 | 0.12 | 6 | 0.3 | 0.13 | 11 | 1.9 | 0.12 |
| Plug-in | 4 | 0.0 | 0.15 | 5 | 0.2 | 0.13 | 11 | 2.0 | 0.12 |
| *Oracles* | | | | | | | | | |
| Prediction oracle | 48 | 8.0 | 0.25 | 29 | 7.3 | 0.15 | 24 | 6.8 | 0.12 |
| Network oracle | 36 | 6.7 | 0.20 | 22 | 6.4 | 0.12 | 20 | 6.5 | 0.11 |

*Note:* Linked designs. True is the percentage of the true nonzero cross-lag coefficients that the fit keeps, False the percentage of the true zero cross-lag coefficients that it keeps, and MCC the Matthews correlation between the kept and the true cross-lag coefficients, set to zero when a fit keeps none. Each true link sits on both lags. Each entry is the mean over the six $N \times T$ cells of a persistence level. Monte Carlo standard errors are at most 0.4 for True and 0.1 for False.

### 4.4 The one-standard-error rule

Moving from $\lambda_{\min}$ to $\lambda_{1se}$ makes every rule shrink more, and whether that helps depends on whether the rule leaked. For random folds it helps. At very high persistence the network error falls from 35 to 3 percent above the oracle, because the heavier penalty partly undoes the overfit. For every rule with a gap it hurts. hv-block with a gap of 6 goes from 4 to 24 percent and the rolling origin from 5 to 42 percent. At low persistence the one-standard-error rule hurts every rule. So $\lambda_{1se}$ with random folds ends up in the middle of the ranking (8.1) because two errors offset each other, and it is beaten by every rule with a gap at $\lambda_{\min}$. Figure 3 puts the two choices side by side.




**(a)** $\lambda_{\min}$ **(b)** $\lambda_{1se}$

**Figure 3:** Error in the connectedness network, by persistence and point on the cross-validation curve

*Note:* Linked designs, $H = 12$. Mean absolute error of the pairwise shares $C_{ij}$, in percent above the network oracle's error in the same cell, at the minimum of the cross-validation curve (a) and at the largest penalty within one standard error of it (b). The horizontal axis is the median first autocorrelation of the simulated series at the low, high and very high persistence levels. Each point is the mean over the six $N \times T$ cells of a level.

Bergmeir et al. (2018) show that random folds are valid for an autoregression when the fitted model leaves no autocorrelation in its residuals. In the simulations this condition fails for almost every penalised fit that recovers the network well, not only for the one-standard-error rule. With the true coefficients, a Ljung–Box test with 12 lags rejects in 6 percent of equations, as a 5 percent test should. At $\lambda_{1se}$ every cross-validation rule leaves autocorrelated residuals in at least 91 percent of equations at high persistence. But hv-block with a gap of 6 at $\lambda_{\min}$ also does so in 49 percent, the rolling origin in 66 and the prediction oracle itself in 34. Only random and interleaved folds at $\lambda_{\min}$ leave nearly white residuals, 13 percent, and they do so because they overfit.[5] White residuals are therefore no guide to the penalty, and the result of Bergmeir et al. (2018) cannot be used to defend random folds for a penalised VAR, because the fits it covers are the overfitted ones.

[5]See Table A.5.

### 4.5 Information criteria and the plug-in penalty

EBIC and the plug-in penalty never use the data to check a forecast, so they cannot leak, but they over-shrink. At very high persistence they add 12 and 16 percent to the network error, and they rank 10.4 and 13.3. BIC does well when every equation has fewer regressors than observations. When $N = 100$ and $T = 120$ it does not, because a fit with as many coefficients as observations has a residual sum of squares close to zero, and its logarithm outweighs BIC's penalty. In these cells BIC keeps 170 cross-lag coefficients per equation, chooses the smallest penalty on the path in 75 percent of equations and gives an explosive VAR in 100 percent of samples. Hecq et al. (2023) recommend BIC with a lower bound on the penalty for this reason. Following their Remark 8, the bounded version minimises BIC, equation by equation, only over the steps of the path before more than $0.5\,T$ slopes are selected. The bound removes the failure. In the cells with more regressors than observations, bounded BIC keeps 4 cross-lag coefficients per equation and gives no explosive VAR. Where plain BIC works, the bound never binds, so the two choose the same penalty. Bounded BIC is within 2 percent of the network oracle at every persistence level, it ranks 4.6 on average, right behind the three rules with a gap, and its index error at very high persistence, 17.8 points, is the smallest of any feasible rule. It gets there in its own way. It chooses heavier penalties than the best predicting one and keeps only 11 to 17 percent of the true cross-lag coefficients, but it almost never keeps a false one, 0.2 percent at low and 4.2 percent at very high persistence (Table 3). Its Matthews correlation, 0.15 at high persistence, is as high as that of any rule.

### 4.6 Sample length

Longer samples shrink every gap, but they do not fix random folds. For hv-block with a gap of 6 the excess network error falls to 0 percent at $N = 100$ and $T = 600$. For random folds it is 16 percent in the same cell, about as large as in the shortest samples, because a longer sample of a persistent series still puts each validation observation next to training observations.[6]

### 4.7 The covariance term

Every row of Table 2 overstates the index, including the oracles. Even with the true coefficients the index is 15.3 points above the truth, averaged over the cells of a persistence level. The reason is the shock covariance. Every estimated table uses the covariance of the fitted residuals, and with an empty network the true covariance is diagonal while the estimated one is not. In that case the moving-average matrices $\Psi_h$ are diagonal, so in Equation (3) the own-lag terms cancel and

$$\theta_{ij}(H) = \frac{\hat{\sigma}_{ij}^2}{\hat{\sigma}_{ii}\hat{\sigma}_{jj}} = \hat{r}_{ij}^2, \tag{9}$$

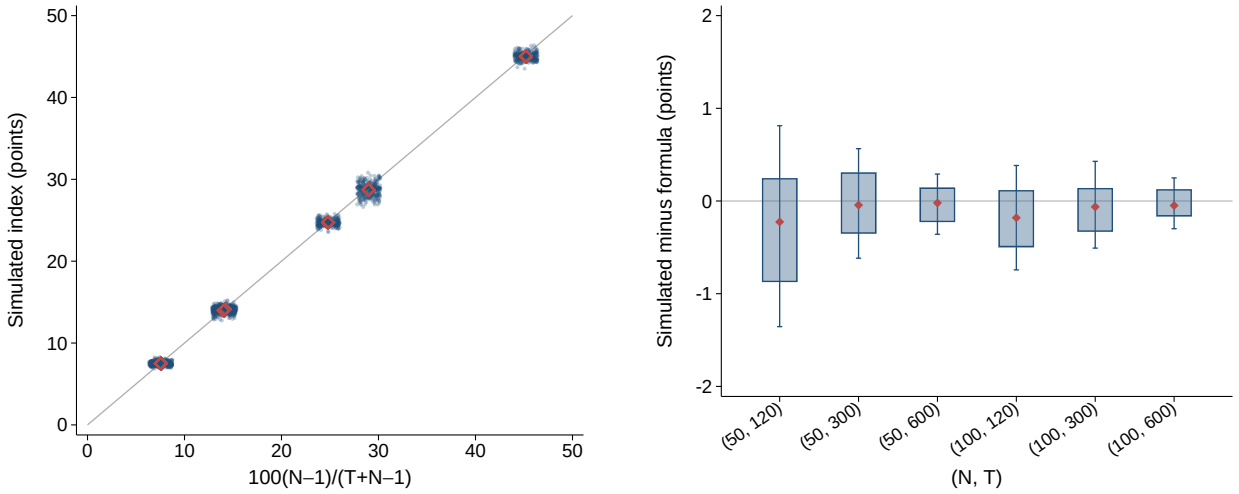
the squared sample correlation between the residuals of $i$ and $j$, at every horizon and whatever the persistence. For independent shocks the squared sample correlation has mean of about $1/T$ (Fisher,

[6] See Table A.3.

1915), so each row has an off-diagonal mass of about $(N-1)/T$ and an own share of one. After the row normalisation in Equation (4),

$$\mathrm{DYCI}_{\mathrm{floor}} \approx 100\,\frac{(N-1)/T}{1+(N-1)/T} = 100\,\frac{N-1}{T+N-1}. \tag{10}$$

Before the normalisation the sum of the squared correlations is the statistic of the Breusch and Pagan (1980) test of independent errors, divided by $T$.




**(a)** Every replication against the formula

**(b)** Simulated minus formula, by cell

**Figure 4:** The connectedness index of an empty network against the formula

*Note:* Designs with no links and uncorrelated shocks. Panel (a) draws one dot per replication, the connectedness index at $H = 12$ computed from the true coefficients and the covariance of their own residuals, against $100(N-1)/(T+N-1)$ for its $N$ and $T$, with a horizontal jitter that spreads them; 300 replications at each of the six $N \times T$ cells, 100 at each persistence level. The hollow diamonds are the cell means and the line is the 45-degree line. Panel (b) draws the same replications as the simulated index minus the formula, by cell. Each box runs from the 25th to the 75th percentile, the whiskers from the 10th to the 90th, and the diamond is the median.

The formula matches the simulations within 0.3 points in every cell (Figure 4), nine replications in ten lie within 1.4 points of it in the smallest cell and within 0.4 in the largest, and the floor is the same at every persistence level, 7.5 points at low and 7.5 at very high persistence for $N = 50$ and $T = 600$. It is large at the sizes applied work uses. At $N = 100$ and $T = 120$ the index of an empty network is 45.0 points, and for 48 series and $T = 120$, a ten-year window of monthly data, it is 28. The first-order approximation $100(N-1)/T$ is good only when $N$ is small relative to $T$. At $N = 100$ and $T = 120$ it gives 82.5.[7]

The tuning rule adds to the floor. With an empty network, random folds add 17.1 points above the floor at very high persistence and hv-block with a gap of 6 adds 7.7 (Table 4). With links, the penalised fits come in below the index of the true coefficients at low persistence, because shrinkage also shrinks some of the residual correlation the floor comes from. So the floor is not simply added

[7] Table A.2 lists both formulas beside the simulated index for every $N \times T$ cell.

to the true index. It sets the scale against which a reported index, and the difference between two tuning rules, should be read.

**Table 4:** Connectedness invented by each rule in an empty network, above the covariance floor

| | Persistence | | |
|---|---|---|---|
| Rule | Low | High | Very high |
| Random 10-fold | 2.2 | 7.0 | 17.1 |
| Interleaved 10-fold | 2.2 | 7.3 | 17.8 |
| Blocked 10-fold | 1.9 | 4.2 | 7.6 |
| hv-block, gap 2 | 1.8 | 4.1 | 7.5 |
| hv-block, gap 6 | 1.7 | 4.2 | 7.7 |
| hv-block, gap 12 | 1.6 | 4.3 | 8.5 |
| Rolling origin | 1.6 | 4.3 | 8.4 |
| Random 10-fold, $\lambda_{1se}$ | 1.7 | 5.3 | 7.6 |
| BIC, bounded | 1.4 | 4.5 | 7.3 |
| EBIC | 3.8 | 6.0 | 11.1 |
| Plug-in | 1.7 | 7.5 | 12.4 |

*Note:* Designs with no links and uncorrelated shocks. Each entry is the estimated connectedness index at $H = 12$ minus the index computed from the true coefficients in the same sample, in points, averaged over the six $N \times T$ cells of a level. Rules at $\lambda_{\min}$ unless marked. Monte Carlo standard errors are at most 0.10.

### 4.8 The one-step table

Every result so far holds at every horizon of two steps or more, and only the one-step table is different.[8] Within each of the 18 cells, the ranking of the 18 feasible rules at $H = 12$ has an average rank correlation of 0.76 with the ranking at $H = 2$, 0.91 at $H = 3$ and at least 0.96 from $H = 5$ on, while at $H = 1$ it is 0.57 and in some cells negative. The best rule at $H = 1$ is one of the three rules that leak, random, interleaved or blocked folds, in 94 percent of the cells. At $H = 2$ the share is 44 percent, at $H = 6$ it is 6 percent, and from $H = 10$ on it is zero. Figure 5 shows the turn for the main contrast. Random folds recover the one-step table slightly better than hv-block with a gap of 6, by 2.8 percent of the oracle's error at very high persistence. From $H = 2$ on they are worse at every persistence level, and the gap widens with the horizon until about ten steps and then stays, at 30.7 percent at very high persistence and $H = 12$.

The reason is in Equation (3). At $H = 1$ the sum has only the term $h = 0$, and $\Psi_0 = I$, so

$$\theta_{ij}(1) = \frac{\hat{\sigma}_{ij}^2}{\hat{\sigma}_{ii}\hat{\sigma}_{jj}} = \hat{r}_{ij}^2. \tag{11}$$

The one-step table is a function of the residual correlations alone, and the lag coefficients, which are what the penalty selects, enter only through the residuals they leave. In the simulations the error of the one-step table follows the error of the residual correlations almost exactly. Across rules within a cell the two have a correlation of 0.99. A heavier penalty leaves more of the lagged dynamics in

[8] Table A.4 reports the excess network error of the main rules at every horizon and persistence level.

the residuals, and two series that are not linked then share part of their unexplained past. At very high persistence, the mean absolute residual correlation between series three or more places apart, whose true correlation is at most 0.03, is 0.051 with the true coefficients, 0.054 with random folds, 0.056 with hv-block and 0.081 with hv-block at $\lambda_{1se}$. Rules that keep more cross-lag coefficients therefore estimate the residual correlations better (correlation $-0.50$ across rules within a cell), and this is the only thing the one-step table rewards. From $H = 2$ on, the table also depends on the lag coefficients through $\Psi_h$, and the false links that random folds keep start to count. So the penalty that gives the best one-step table is not the penalty that gives the best table at any longer horizon, and a one-step table should not be used to choose a tuning rule.

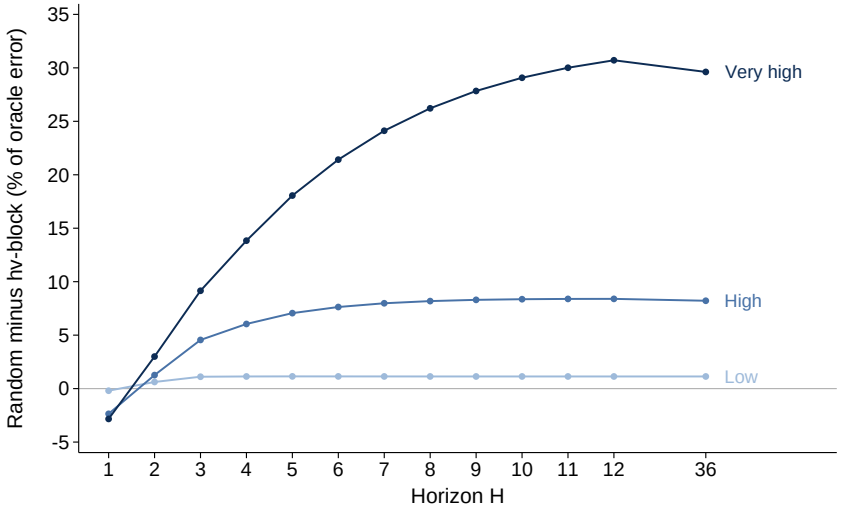



**Figure 5:** Random folds against hv-block with a gap of 6, by forecast horizon

*Note:* Linked designs. Network error of random ten-fold cross-validation minus that of hv-block cross-validation with a gap of 6 observations, both at $\lambda_{\min}$, in percent of the network oracle's error at the same horizon, for the three persistence levels, at $H = 1, \ldots, 12$ and 36 (the last placed after 12 on the axis). Positive values mean random folds recover the network worse. Each point is the mean over the six $N \times T$ cells of a level.

## 5 Conclusion

When the series are persistent, cross-validation with random folds overfits an elastic-net VAR. It promises a forecast it does not deliver, keeps links that are not there and raises the connectedness index. The fix is simple and costs nothing. Choosing the penalty at the minimum of the cross-validation curve, with a gap of 6 to 12 observations between training and validation or with a rolling origin, comes within a few percent of the best penalty for the network at every persistence level I consider. BIC with a lower bound on the penalty does as well and needs no cross-validation. The one-standard-error rule, EBIC and the plug-in penalty over-shrink, and plain BIC fails when an equation has more regressors than observations. A researcher should also report $100(N-1)/(T+N-1)$ next to the index, the part of it that noise in the estimated covariance alone produces, because that part can be large whatever the penalty.

The designs are sparse VARs with a known network and no common factor. Housing and financial series share a strong common component, and the next step is to place the estimated index of monthly state house prices in this grid and to ask how the ranking changes when a factor drives part of the persistence.

## References


Belloni, A., Chen, D., Chernozhukov, V., and Hansen, C. (2012). Sparse models and methods for optimal instruments with an application to eminent domain. *Econometrica*, 80(6):2369–2429.

Bergmeir, C., Hyndman, R. J., and Koo, B. (2018). A note on the validity of cross-validation for evaluating autoregressive time series prediction. *Computational Statistics & Data Analysis*, 120:70–83.

Bonaccolto, G., Caporin, M., and Shahzad, S. J. H. (2026). (Quantile) spillover indexes: Simulation-based evidence, confidence intervals and a decomposition. *Journal of Financial Econometrics*, 24(1).

Bostancı, G. and Yılmaz, K. (2020). How connected is the global sovereign credit risk network? *Journal of Banking & Finance*, 113:105761.

Breusch, T. S. and Pagan, A. R. (1980). The Lagrange multiplier test and its applications to model specification in econometrics. *Review of Economic Studies*, 47(1):239–253.

Brunner, F. and Hipp, R. (2023). Estimating large-dimensional connectedness tables: The great moderation through the lens of sectoral spillovers. *Quantitative Economics*, 14(3):1021–1058.

Burman, P., Chow, E., and Nolan, D. (1994). A cross-validatory method for dependent data. *Biometrika*, 81(2):351–358.

Chen, J. and Chen, Z. (2008). Extended Bayesian information criteria for model selection with large model spaces. *Biometrika*, 95(3):759–771.

Demirer, M., Diebold, F. X., Liu, L., and Yilmaz, K. (2018). Estimating global bank network connectedness. *Journal of Applied Econometrics*, 33(1):1–15.

Diebold, F. X., Liu, L., and Yilmaz, K. (2017). Commodity connectedness. Working Paper 23685, National Bureau of Economic Research.

Diebold, F. X. and Yilmaz, K. (2012). Better to give than to receive: Predictive directional measurement of volatility spillovers. *International Journal of Forecasting*, 28(1):57–66.

Diebold, F. X. and Yilmaz, K. (2014). On the network topology of variance decompositions: Measuring the connectedness of financial firms. *Journal of Econometrics*, 182(1):119–134.

Fisher, R. A. (1915). Frequency distribution of the values of the correlation coefficient in samples from an indefinitely large population. *Biometrika*, 10(4):507–521.

Friedman, J., Hastie, T., and Tibshirani, R. (2010). Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software*, 33(1):1–22.

Gabauer, D. (2025). ConnectednessApproach: Connectedness approach, R package version 1.0.4. CRAN.

Gabauer, D., Gupta, R., Marfatia, H. A., and Miller, S. M. (2024). Estimating U.S. housing price network connectedness: Evidence from dynamic elastic net, lasso, and ridge vector autoregressive models. *International Review of Economics & Finance*, 89:349–362.

Greenwood-Nimmo, M., Kočenda, E., and Nguyen, V. H. (2024). Detecting statistically significant changes in connectedness: A bootstrap-based technique. *Economic Modelling*, 140:106843.

Greenwood-Nimmo, M. and Tarassow, A. (2022). Bootstrap-based probabilistic analysis of spillover scenarios in economic and financial networks. *Journal of Financial Markets*, 59:100661.

Gross, C. and Siklos, P. L. (2020). Analyzing credit risk transmission to the nonfinancial sector in Europe: A network approach. *Journal of Applied Econometrics*, 35(1):61–81.

Hecq, A., Margaritella, L., and Smeekes, S. (2023). Granger causality testing in high-dimensional VARs: A post-double-selection procedure. *Journal of Financial Econometrics*, 21(3):915–958.

Jiménez-Varón, C. F. and Knight, M. I. (2025). Frequency-domain analysis of time series with network-structured dependence: Application to global bank connectedness. arXiv:2510.06157.

Koop, G., Pesaran, M. H., and Potter, S. M. (1996). Impulse response analysis in nonlinear multivariate models. *Journal of Econometrics*, 74(1):119–147.

Lusompa, A. (2025). A note on the finite sample bias in time series cross-validation. arXiv:2512.05900.

Manso, J. (2026). Volatility spillovers in China's real estate crisis: A network approach. arXiv:2602.19740.

Nicholson, W. B., Wilms, I., Bien, J., and Matteson, D. S. (2020). High dimensional forecasting via interpretable vector autoregression. *Journal of Machine Learning Research*, 21(166):1–52.

Pesaran, H. H. and Shin, Y. (1998). Generalized impulse response analysis in linear multivariate models. *Economics Letters*, 58(1):17–29.

Racine, J. (2000). Consistent cross-validatory model-selection for dependent data: hv-block cross-validation. *Journal of Econometrics*, 99(1):39–61.

Schwarz, G. (1978). Estimating the dimension of a model. *Annals of Statistics*, 6(2):461–464.

Zou, H. and Hastie, T. (2005). Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society Series B*, 67(2):301–320.

# A Additional tables

**Table A.1:** What each cross-validation rule promises and what it chooses, by persistence

| | Persistence | | |
|---|---|---|---|
| Rule | Low | High | Very high |
| *A. Optimism (cross-validated minus true error)* | | | |
| Random 10-fold | −0.040 | −0.202 | −1.263 |
| Interleaved 10-fold | −0.041 | −0.213 | −1.305 |
| Blocked 10-fold | −0.024 | −0.058 | −0.479 |
| hv-block, gap 2 | −0.021 | −0.041 | −0.412 |
| hv-block, gap 6 | −0.017 | −0.015 | −0.274 |
| hv-block, gap 12 | −0.011 | 0.010 | −0.120 |
| Rolling origin | −0.006 | 0.015 | −0.194 |
| *B. Path steps of less shrinkage than the prediction oracle* | | | |
| Random 10-fold | 0.7 | 4.0 | 10.9 |
| Interleaved 10-fold | 0.7 | 4.3 | 11.4 |
| Blocked 10-fold | −0.1 | 0.0 | 1.7 |
| hv-block, gap 2 | −0.1 | −0.4 | 1.1 |
| hv-block, gap 6 | −0.4 | −0.9 | −0.1 |
| hv-block, gap 12 | −0.7 | −1.4 | −1.2 |
| Rolling origin | −1.6 | −2.5 | −1.9 |
| Network oracle | −1.1 | −1.0 | 0.0 |
| *C. Cross-lag coefficients kept per equation* | | | |
| Random 10-fold | 13.8 | 20.5 | 30.1 |
| Interleaved 10-fold | 14.0 | 21.3 | 31.5 |
| Blocked 10-fold | 11.9 | 10.4 | 11.6 |
| hv-block, gap 2 | 11.7 | 9.7 | 10.8 |
| hv-block, gap 6 | 11.3 | 8.9 | 9.5 |
| hv-block, gap 12 | 10.6 | 8.2 | 8.5 |
| Rolling origin | 8.5 | 6.6 | 8.0 |
| Prediction oracle | 12.6 | 10.7 | 10.0 |
| Network oracle | 10.6 | 10.0 | 9.9 |

*Note:* Linked designs, $\lambda_{\min}$ of each rule. Each entry is the mean over the six $N \times T$ cells of a persistence level, each cell weighted equally. Optimism is the cross-validated mean squared error at the chosen step minus the true out-of-sample error at that step, averaged over equations, in units of the shock variance. Path steps count the steps of the 100-step penalty path between the chosen penalty and the prediction oracle's, averaged over equations; a positive number means less shrinkage. Each step changes the penalty by about 10 percent. Cross-lag coefficients are the nonzero coefficients on other series' lags per equation; the true VAR has 2.5 links per equation, each on both lags, so 5 nonzero cross-lag coefficients. All seven cross-validation rules; Figure 1 shows four of them. Monte Carlo standard errors are at most 0.009 in panel A, 0.04 in panel B and 0.08 in panel C.

**Table A.2:** The connectedness index of an empty network

| $N$ | $T$ | Simulated | $100(N-1)/(T+N-1)$ | $100(N-1)/T$ |
|---|---|---|---|---|
| 50 | 120 | 28.7 | 29.0 | 40.8 |
| 50 | 300 | 14.0 | 14.0 | 16.3 |
| 50 | 600 | 7.5 | 7.6 | 8.2 |
| 100 | 120 | 45.0 | 45.2 | 82.5 |
| 100 | 300 | 24.7 | 24.8 | 33.0 |
| 100 | 600 | 14.1 | 14.2 | 16.5 |

*Note:* Designs with no links and uncorrelated shocks, so the true index is zero. Simulated is the mean connectedness index at $H = 12$ computed from the true coefficients and the covariance of their own residuals, pooled over the three persistence levels; Monte Carlo standard errors are at most 0.05. The last two columns are the floor in Equation (10) and its first-order approximation.

**Table A.3:** Excess network error by the number of series and the length of the sample

| | $N = 50$ | | | $N = 100$ | | |
|---|---|---|---|---|---|---|
| Rule | $T = 120$ | $T = 300$ | $T = 600$ | $T = 120$ | $T = 300$ | $T = 600$ |
| Random 10-fold | 15 | 18 | 13 | 13 | 17 | 16 |
| Blocked 10-fold | 5 | 3 | 2 | 3 | 1 | 0 |
| hv-block, gap 6 | 3 | 2 | 2 | 4 | 0 | 0 |
| hv-block, gap 12 | 4 | 1 | 1 | 6 | 0 | −1 |
| Rolling origin | 5 | 1 | 0 | 6 | 2 | −1 |
| Random 10-fold, $\lambda_{1se}$ | 5 | 4 | 4 | 4 | 4 | 4 |
| BIC | 3 | 1 | 1 | 89 | 2 | 1 |
| BIC, bounded | 3 | 1 | 1 | 4 | 2 | 1 |
| EBIC | 12 | 5 | 3 | 16 | 8 | 4 |
| Plug-in | 17 | 9 | 7 | 15 | 11 | 7 |

*Note:* Linked designs, $H = 12$. Mean absolute error of the pairwise shares in percent above the network oracle's, averaged over the three persistence levels. Rules at $\lambda_{\min}$ unless marked.

**Table A.4:** Excess network error by forecast horizon and persistence

| | $\lambda_{\min}$ | | | | $\lambda_{1se}$ | | Information criteria | |
|---|---|---|---|---|---|---|---|---|
| $H$ | Random | Blocked | hv, gap 6 | Rolling | Random | hv, gap 6 | BIC, bounded | EBIC |
| *A. Low persistence* | | | | | | | | |
| 1 | -1 | -1 | -1 | 0 | 6 | 7 | 4 | 7 |
| 2 | 1 | 0 | 0 | 0 | 5 | 5 | 3 | 6 |
| 3 | 1 | 1 | 0 | 0 | 4 | 4 | 2 | 5 |
| 4 | 1 | 1 | 0 | 0 | 4 | 4 | 2 | 5 |
| 6 | 1 | 1 | 0 | 0 | 4 | 4 | 2 | 5 |
| 9 | 1 | 1 | 0 | 0 | 4 | 4 | 2 | 5 |
| 12 | 1 | 0 | 0 | 0 | 4 | 4 | 2 | 5 |
| 36 | 1 | 0 | 0 | 0 | 4 | 4 | 2 | 5 |
| *B. High persistence* | | | | | | | | |
| 1 | -2 | -1 | 0 | 2 | 11 | 23 | 7 | 15 |
| 2 | 2 | 0 | 0 | 2 | 9 | 21 | 5 | 12 |
| 3 | 5 | 0 | 1 | 2 | 7 | 18 | 4 | 10 |
| 4 | 7 | 1 | 1 | 1 | 7 | 16 | 3 | 9 |
| 6 | 9 | 2 | 1 | 1 | 6 | 15 | 3 | 8 |
| 9 | 10 | 2 | 2 | 1 | 6 | 14 | 2 | 8 |
| 12 | 10 | 2 | 2 | 1 | 5 | 14 | 2 | 8 |
| 36 | 10 | 2 | 1 | 1 | 5 | 14 | 2 | 7 |
| *C. Very high persistence* | | | | | | | | |
| 1 | -1 | -2 | 1 | 7 | 5 | 45 | 3 | 24 |
| 2 | 5 | -1 | 2 | 7 | 6 | 43 | 3 | 23 |
| 3 | 11 | 0 | 2 | 8 | 6 | 40 | 3 | 21 |
| 4 | 16 | 1 | 2 | 7 | 6 | 37 | 3 | 19 |
| 6 | 25 | 3 | 3 | 7 | 5 | 32 | 3 | 16 |
| 9 | 32 | 4 | 4 | 6 | 4 | 27 | 2 | 13 |
| 12 | 35 | 4 | 4 | 5 | 3 | 24 | 2 | 12 |
| 36 | 33 | 4 | 3 | 4 | 2 | 18 | 1 | 9 |

*Note:* Linked designs. Mean absolute error of the pairwise shares at horizon $H$, in percent above the network oracle's in the same cell and replication, averaged over the six $N \times T$ cells of a persistence level. The network oracle is chosen at $H = 12$ and evaluated at each horizon, so at other horizons a rule can beat it.

**Table A.5:** Residual autocorrelation of the selected fits

| | Ljung–Box rejections (%) | | | Median \|ACF(1)\| | | |
|---|---|---|---|---|---|---|
| Rule | Low | High | Very high | Low | High | Very high |
| Random 10-fold | 14 | 13 | 7 | 0.09 | 0.07 | 0.04 |
| Interleaved 10-fold | 14 | 12 | 7 | 0.09 | 0.07 | 0.04 |
| Blocked 10-fold | 17 | 41 | 44 | 0.10 | 0.13 | 0.12 |
| hv-block, gap 2 | 18 | 44 | 47 | 0.10 | 0.13 | 0.13 |
| hv-block, gap 6 | 18 | 49 | 58 | 0.10 | 0.14 | 0.15 |
| hv-block, gap 12 | 20 | 54 | 67 | 0.11 | 0.16 | 0.18 |
| Rolling origin | 26 | 66 | 74 | 0.12 | 0.17 | 0.18 |
| Random 10-fold, $\lambda_{1se}$ | 65 | 91 | 75 | 0.19 | 0.23 | 0.17 |
| Blocked 10-fold, $\lambda_{1se}$ | 66 | 98 | 97 | 0.19 | 0.29 | 0.29 |
| hv-block, gap 6, $\lambda_{1se}$ | 66 | 99 | 99 | 0.19 | 0.31 | 0.34 |
| Rolling origin, $\lambda_{1se}$ | 68 | 99 | 99 | 0.21 | 0.36 | 0.41 |
| BIC | 57 | 90 | 79 | 0.17 | 0.21 | 0.19 |
| BIC, bounded | 57 | 88 | 72 | 0.17 | 0.20 | 0.16 |
| EBIC | 66 | 98 | 95 | 0.19 | 0.25 | 0.25 |
| Plug-in | 69 | 100 | 100 | 0.21 | 0.29 | 0.29 |
| Prediction oracle | 13 | 34 | 54 | 0.10 | 0.12 | 0.15 |
| Network oracle | 31 | 48 | 55 | 0.12 | 0.12 | 0.13 |
| True coefficients | 6 | 6 | 6 | 0.04 | 0.04 | 0.04 |

*Note:* Linked designs. Residuals of each selected fit on its estimation sample, from the simulated histories regenerated with their stored seeds (the regenerated residuals reproduce every stored residual covariance to 6e-15). Ljung–Box rejections is the percentage of equations whose Ljung–Box test with 12 lags rejects no autocorrelation at 5 percent. Median |ACF(1)| is the median over equations of the absolute first autocorrelation of the residuals. Each entry is the mean over the six $N \times T$ cells of a persistence level. Rules at $\lambda_{\min}$ unless marked.